00-M-251C
April 20, 1971
SUPERSEDING
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May 3, 1962

FEDERAL SPECIFICATION

MANDRELS, MACHINE, SOLID AND EXPANSION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers solid and expansion-type mandrels used for supporting work pieces between centers for machining purposes and inspection operations.

1.2 Classification.

1.2.1 Types and styles. The mandrels covered by this specification shall be of the following types and styles as specified (see 6.2):

Type I. - Solid (see Figure 1).

Type II. - Expanding sleeve type, rectangular slit (see Figure 2) or drill sleeve type (see Figure 3).

Style I. - Fast taper, light duty (see Table II).
Style II. - Medium taper, medium duty (see Table II).
Style III. - Slow taper, heavy duty (see Table II).

Type III. - Expanding jaw types (see Figures 4 and 5).

Style I. - Step jaw (see Table III).
Style II. - Straight jaw (see Table IV).

FSC 3460

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1.2.2 Sizes. The mandrels covered by this specification shall be of the sizes specified herein (see Tables I, II, III, and IV), as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-T-1150	-	Tools and Tool Accessories For Power Driven, Metal and Woodworking, Machinery; Packaging and Packing of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of Military Specifications, Standards, Drawings, and Publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute, Inc.

B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish one complete mandrel for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The mandrels shall be in substantial agreement with the figures and tables to which reference is made, including the requirements of 3.7, 3.8, and 3.9.

3.2.1 Illustrations. The illustrations shown herein (see Figures 1 to 5, inclusive) are for the convenience of identification and are not intended to preclude the purchase of mandrels which are otherwise in accordance with this specification.

3.3 Material. High-grade carbon or alloy steel shall be used for the manufacture of the mandrels. The steel shall be entirely suitable for the intended purpose.

3.4 Construction. The mandrels shall be free from any characteristics or defects that may render the mandrels unsuitable for the purpose intended. All sharp edges shall be broken. In addition, the two ends of type I mandrel body shall be rounded as specified in Table I. The radii of the types II and III mandrel body ends shall conform to the manufacturer's practice.

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TABLE I. Type I, solid, dimensions in inches

Intermediate sizes take the linear dimensions of the next larger size.

Nom size	Overall length	Body length	Neck length	Neck dia	Large dia	Small dia	Entering chamfer	Length of flat	Distance across flat	Radius or chamfer	Center Hole	
											Drill dia	C'sink dia
A	B	C	D	E	F	G	H	J	K	L	M	N
1/8	2 1/2	1 7/8	5/16	7/64	0.1265	0.1247	1/32	1/4	0.100	1/64	0.024	0.050
5/32	3	2 1/4	3/8	9/64	.1578	.1560	1/32	5/16	.129	1/64	.028	.075
3/16	3	2 1/4	3/8	11/64	.1890	.1872	1/32	5/16	.158	1/64	.028	.100
7/32	3 3/4	2 7/8	7/16	13/64	.2202	.2184	1/32	5/16	.172	1/64	3/64	3/32
1/4	3 3/4	2 7/8	7/16	7/32	.2515	.2495	1/32	5/16	.202	1/64	3/64	3/32
9/32	4	3	1/2	1/4	.2828	.2808	1/32	3/8	.236	1/64	3/64	3/32
5/16	4	3	1/2	9/32	.3140	.3120	1/32	3/8	.267	1/64	3/64	3/32
11/32	4 1/4	3 1/4	1/2	5/16	.3452	.3432	1/32	3/8	.297	1/32	3/64	3/32
3/8	4 1/4	3 1/4	1/2	11/32	.3765	.3745	1/32	3/8	.327	1/32	1/16	5/32
13/32	4 1/2	3 3/8	9/16	3/8	.4080	.4058	1/32	7/16	.356	1/32	1/16	5/32
7/16	4 1/2	3 3/8	9/16	13/32	.4392	.4370	1/32	7/16	.386	1/32	1/16	5/32
15/32	5	3 3/4	5/8	7/16	.4704	.4682	1/32	1/2	.416	1/32	1/16	5/32
1/2	5	3 3/4	5/8	15/32	.5017	.4995	1/32	1/2	.441	1/32	5/64	13/64
9/16	5 1/4	4	5/8	33/64	.5644	.5620	1/16	1/2	.485	1/16	5/64	13/64
5/8	5 1/2	4	3/4	37/64	.6269	.6247	1/16	5/8	.547	1/16	5/64	13/64
11/16	5 3/4	4 1/4	3/4	41/64	.6894	.6870	1/16	5/8	.608	1/16	3/32	7/32
3/4	6	4 1/2	3/4	11/16	.7522	.7495	1/16	5/8	.653	1/16	3/32	7/32
13/16	6 1/4	4 1/2	7/8	3/4	.8147	.8120	1/16	3/4	.713	1/16	3/32	7/32
7/8	6 1/2	4 3/4	7/8	13/16	.8772	.8745	1/16	3/4	.772	1/16	3/32	7/32
15/16	6 3/4	5	7/8	7/8	.9397	.9370	1/16	3/4	.896	3/32	3/32	7/32
1	7	5	1	15/16	1.0022	.9995	1/16	3/4	.896	3/32	5/32	11/32
1 1/16	7 1/4	5 1/4	1	15/16	1.0650	1.0620	3/32	3/4	.900	3/32	5/32	11/32
1 1/8	7 1/2	5 1/2	1	1	1.1275	1.1245	3/32	3/4	.960	3/32	5/32	11/32
1 3/16	7 3/4	5 1/2	1 1/8	1 1/16	1.1900	1.1870	3/32	7/8	1.020	3/32	5/32	11/32
1 1/4	8	5 3/4	1 1/8	1 1/8	1.2528	1.2490	3/32	7/8	1.080	3/32	7/32	13/32
1 5/16	8 1/4	6	1 1/8	1 3/16	1.3153	1.3115	3/32	7/8	1.140	3/32	7/32	3/8
1 3/8	8 1/2	6	1 1/4	1 1/4	1.3778	1.3740	3/32	1	1.200	3/32	7/32	3/8
1 7/16	8 3/4	6 1/4	1 1/4	1 5/16	1.4403	1.4365	3/32	1	1.260	3/32	7/32	3/8
1 1/2	9	6 1/2	1 1/4	1 3/8	1.5028	1.4990	3/32	1	1.380	3/32	7/32	3/8
1 9/16	9 1/4	6 1/2	1 3/8	1 3/8	*	1.5617	3/32	1 1/8	1.320	1/8	7/32	3/8
1 5/8	9 1/2	6 3/4	1 3/8	1 7/16	1.6279	1.6240	3/32	1 1/8	1.380	1/8	7/32	3/8
1 11/16	9 3/4	7	1 3/8	1 1/2	*	1.6867	3/32	1 1/8	1.440	1/8	7/32	3/8
1 3/4	10	7 1/4	1 3/8	1 9/16	1.7529	1.7490	3/32	1 1/8	1.500	1/8	7/32	3/8
1 13/16	10 1/4	7 1/4	1 1/2	1 5/8	*	1.8117	3/32	1 1/4	1.560	1/8	7/32	3/8
1 7/8	10 1/2	7 1/2	1 1/2	1 11/16	1.8780	1.8740	3/32	1 1/4	1.620	1/8	7/32	3/8
1 15/16	10 3/4	7 3/4	1 1/2	1 3/4	*	1.9367	3/32	1 1/4	1.680	1/8	7/32	3/8
2	11	8	1 1/2	1 3/4	2.0035	1.9990	1/8	1 1/4	1.680	1/8	7/32	1/2
2 1/8	11 1/2	8 1/4	1 5/8	1 7/8	*	2.1240	1/8	1 1/4	1.800	1/8	7/32	1/2
2 1/4	12	8 3/4	1 5/8	2	*	2.2490	1/8	1 1/4	1.920	1/8	7/32	1/2
2 3/8	12	8 3/4	1 5/8	2 1/8	*	2.3740	1/8	1 1/4	2.040	1/8	7/32	1/2
2 1/2	12 1/2	9 1/4	1 5/8	2 1/4	*	2.4990	1/8	1 1/4	2.160	1/8	7/32	1/2
2 5/8	12 1/2	9 1/4	1 5/8	2 3/8	*	2.6240	1/8	1 1/4	2.280	1/8	7/32	1/2
2 3/4	13	9 1/2	1 3/4	2 1/2	*	2.7490	1/8	1 3/8	2.400	1/8	7/32	1/2
2 7/8	13	9 1/2	1 3/4	2 5/8	*	2.8740	1/8	1 3/8	2.525	1/8	7/32	1/2
3	13	9 1/2	1 3/4	2 3/4	*	2.9990	1/8	1 3/8	2.640	1/8	1/4	39/64
3 1/4	14	10 1/4	1 7/8	2 3/4	*	3.2490	1/8	1 1/2	2.640	1/8	1/4	39/64
3 1/2	15	11 1/4	1 7/8	3	*	3.4990	1/8	1 1/2	2.890	1/8	1/4	39/64
3 3/4	16	12	2	3 1/4	*	3.7490	1/8	1 5/8	3.140	1/8	1/4	39/64
4	17	12 3/4	2 1/8	3 1/2	*	3.9990	1/8	1 5/8	3.390	1/8	1/4	39/64
	±1/16 inch		±1/32 inch		*0.0005 taper per in. (length)				±0.005 thru 1 1/2 nom size ±0.008 over 1 1/2 nom size			

	MANDREL SIZE		
	Thru 1 1/2	Over 1 1/2 thru 1 7/8	Over 1 7/8
Diameter Tolerance Large End (F)	+.0005	+.0006	+.0007
Diameter Tolerance Small End (G)	-.0003	-.0003	-.0003

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3.5 Interchangeability. All replaceable parts of the expanding-type mandrels shall be constructed to definite tolerances and clearances in order that any such part of a particular type or size may be replaced or adjusted without requiring modification.

3.6 Identification marking. Each mandrel shall be permanently marked in accordance with the manufacturer's standard commercial practice, including the size.

3.7 Type I, solid mandrel.

3.7.1 Design. The solid mandrel shall conform to Figure 1 and to the requirements of Table I.

3.7.2 Hardness. The finished mandrel shall have a hardness value of not less than 60 nor more than 65 on the Rockwell "C" scale on all surfaces.

3.7.3 Tolerances and finishes. Finish grinding, honing, or lapping of mandrels shall be done after hardening and stress relieving. Mandrel centers shall be accurately ground and lapped to the standard included angle of 60 degrees ± 15 minutes. Mandrels shall be ground on dead centers to a uniform straight-taper surface to the dimensions listed in Table I for each size of mandrel. The surface roughness of the finish shall not exceed 16 microinches arithmetical average (a.a.), and shall also conform to B46.1. The maximum allowable runout for all sizes of mandrels shall not exceed 0.0005 inch total indicator reading at any position along the working surface when rotated on snugly adjusted dead centers.

3.8 Type II, expanding sleeve type, rectangular slit or drill sleeve type, styles I, II, and III.

3.8.1 Design. The expanding mandrel shall be in substantial conformance with Figures 2 and 3. Mandrels shall consist of a solid arbor finished to dimensions listed in Table II and one or more, one-piece, split, flexible, spring-tempered sleeves. Sleeves shall be finished internally to fit the arbor. Sleeves shall also be so proportioned as to provide uniform expansion, without failure, to any size in the range of diameters listed in Table II for each sleeve. Each sleeve shall have an even number of slits one-half of which shall open at each end of the sleeve. Each slit shall terminate in a drilled hole near the end of the sleeve. The square slits shall have fillets of at least 1/64 inch. The diameter of the hole shall be approximately three times the width of the slit. The number of slits in relation to the diameter and thickness of a sleeve shall be such as to provide adequate strength and flexibility, and suitable bearing on the arbor and in the work hole for all sizes in the range for each sleeve.

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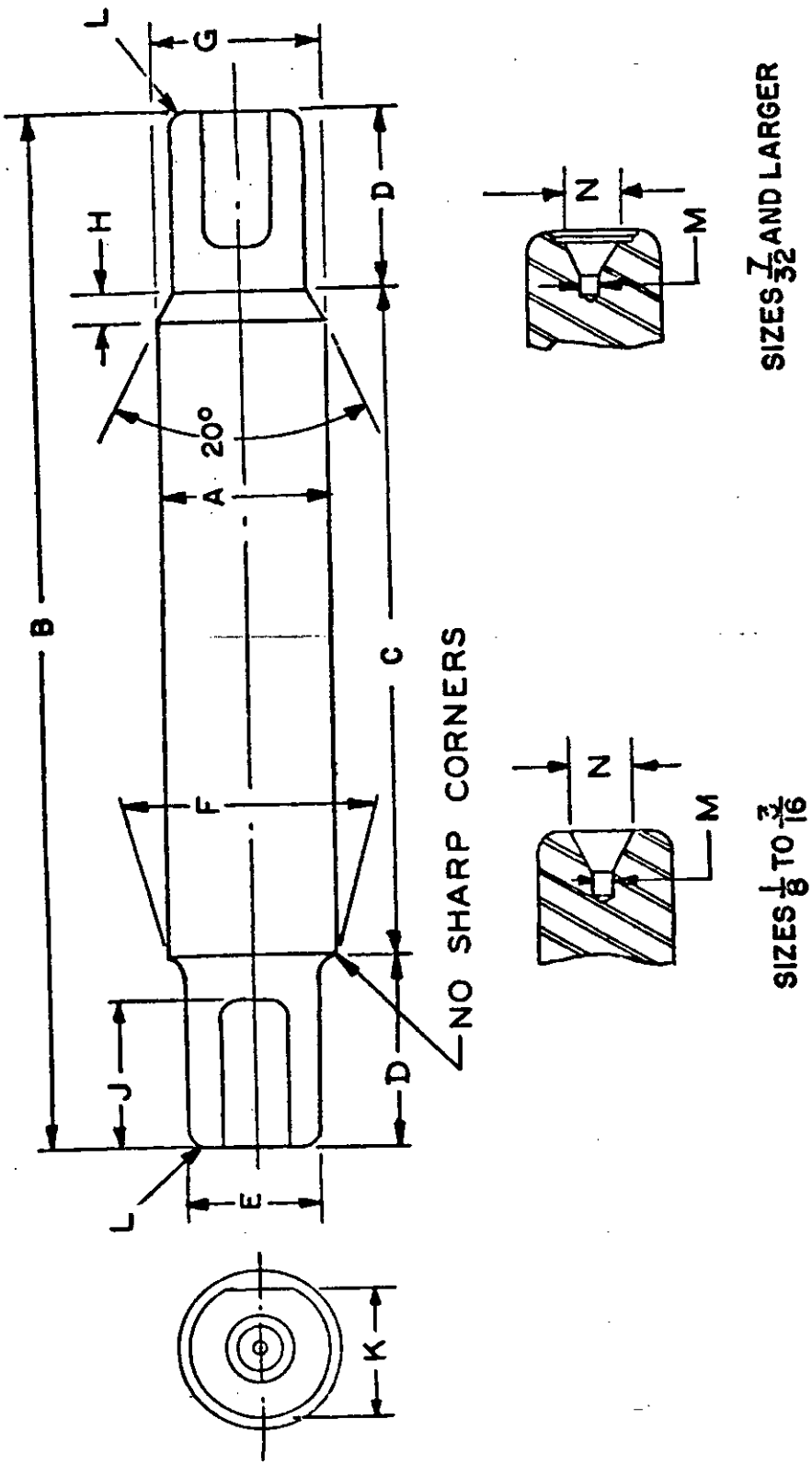


FIGURE 1.—Type I, solid mandrel.

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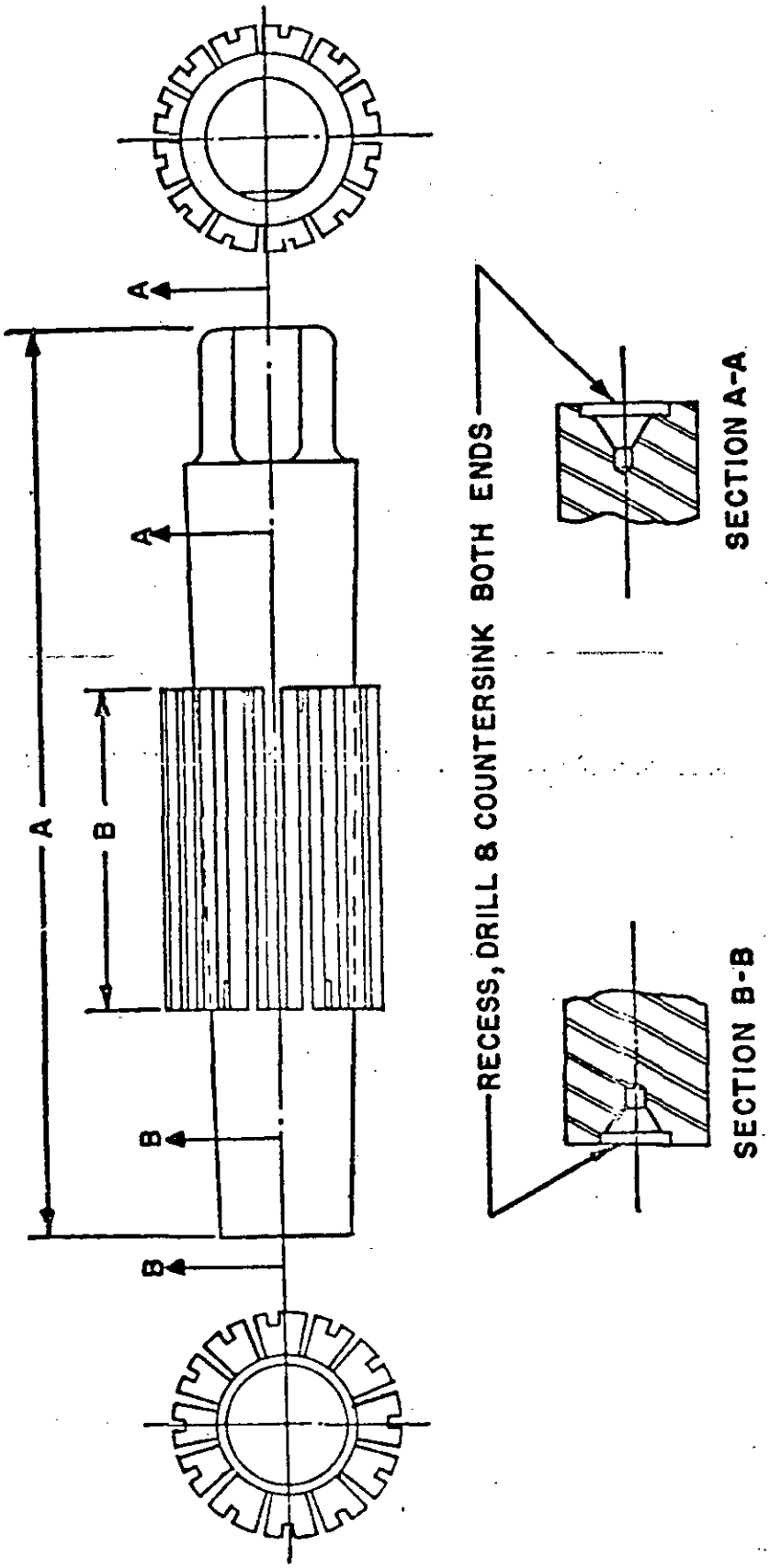


FIGURE 2.—Type II, expanding sleeve, rectangular slit.

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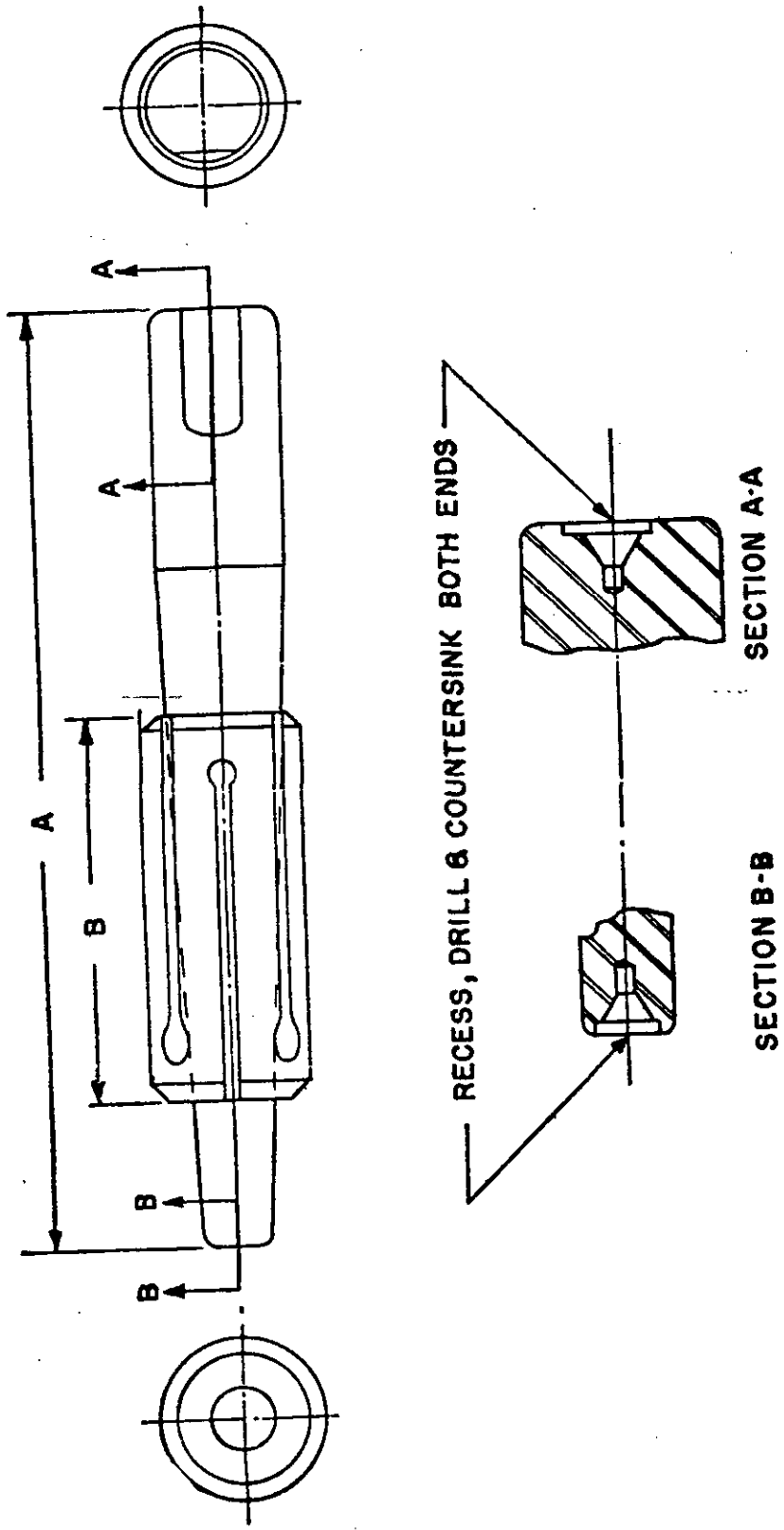


FIGURE 8.—Type II, expanding, drill-type sleeve.

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TABLE II. Type II, styles I, II, and III expanding sleeve type, rectangular slit or drill sleeve type

Style I, fast taper, light duty 1 1/8 inches per foot taper				Style II, medium taper, medium duty minimum 3/4 inch per foot taper				Style III, slow taper, heavy duty 1/2 inch per foot taper			
Range, inclusive		Arbor length	Sleeve length	Range, inclusive		Arbor length	Sleeve length	Range, inclusive		Arbor length	Sleeve length
Minimum	Maximum	A +0.005 - .005	B +0.005 - .005	Minimum	Maximum	A +0.005 - .005	B +0.005 - .005	Minimum	Maximum	A +0.005 - .005	B +0.005 - .005
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1 9/16	2 1/2	16 3/4	4 1/4	3/8	1/2	4 1/2	1 1/4	11/16	27/32	8	2 1/4
2 1/2	4	17 3/4	7	1/2	9/16	5	2 1/2	27/32	1	7	2 1/2
2 1/2	5 1/2	18	7	1/2	11/16	5 3/4	1 1/2	1	1 9/32	12 1/4	3 1/4
3 1/4	4	18	7	9/16	21/32	6	2 3/4	1 9/32	1 9/16	12 1/4	3 1/4
3 1/4	4 3/4	18	7	21/32	3/4	6	2 3/4	.....	.....	.....	.....
4	4 3/4	18	7	11/16	1	9 1/4	2 1/2	.....	.....	.....	.....
4	5 1/2	18	7	3/4	7/8	7	3 1/4	.....	.....	.....	.....
4 3/4	5 1/2	18 1/4	7	13/16	1	9 1/4	2 1/2	.....	.....	.....	.....
.....	.....	.....	.....	7/8	1	7	3 1/4	.....	.....	.....	.....
.....	.....	.....	.....	1	1 3/8	9	4	.....	.....	.....	.....
.....	.....	.....	.....	1	1 9/16	12 1/4	3 1/4	.....	.....	.....	.....
.....	.....	.....	.....	1 1/4	1 5/8	9	4	.....	.....	.....	.....
.....	.....	.....	.....	1 1/4	1 5/8	9	4 1/4	.....	.....	.....	.....
.....	.....	.....	.....	1 1/2	2	11 1/2	5	.....	.....	.....	.....
.....	.....	.....	.....	2	2 3/4	14	5 1/2	.....	.....	.....	.....
.....	.....	.....	.....	2 3/4	3 3/4	17	6	.....	.....	.....	.....
.....	.....	.....	.....	3 3/4	5	20	7	.....	.....	.....	.....
.....	.....	.....	.....	5	6 1/2	24	8	.....	.....	.....	.....

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3.8.2 Hardness. The sleeve shall have a hardness of not less than 42 nor more than 45, and the arbors shall have a hardness of not less than 56 nor more than 58 on the Rockwell "C" scale on all surfaces.

3.8.3 Tolerances and finishes. Finish grinding, honing, or lapping of arbors and sleeves shall be done after hardening and stress relieving. Arbor centers shall be accurately ground and lapped to the standard included angle of 60 degrees ± 30 minutes. Arbors shall be ground on dead centers to a uniform straight-taper surface. The surface roughness of arbors and the external and internal roughness of the sleeves shall not exceed 16 microinches a.a. and shall also conform to B46.1. Expanding mandrels shall have a runout not exceeding 0.002 inch total indicator reading at any position along the working surface of the sleeve for any position of the sleeve on the arbor within the size range for each sleeve and arbor.

3.8.4 Repair parts. When required, additional sleeves for type II mandrels shall be furnished in the sizes and quantities as specified (see 6.2).

3.9 Type III, expanding jaw types, styles I and II.

3.9.1 Design. The jaw-type expanding mandrel shall be in substantial conformance with Figures 4 and 5 and shall consist of an arbor with slots for the sliding jaws and sets of three or four jaws to the dimensions listed in Tables III (step jaws) and IV (straight jaws). The jaw-bearing surfaces under tension shall have a runout not exceeding 0.001 inch total indicator reading at any position along the working surface of the jaw bearings.

TABLE III. Type III, style I, expanding mandrels, jaw type, step jaw.

Range of Expansion		Jaw Bearing Length of Step			Arbor Length, Minimum A	Width of Jaws, Minimum B	Number of Jaws, Minimum
Minimum	Maximum	Small	Intermediate	Large			
Inches	Inches	Inches	Inches	Inches	Inches	Inches	
1/2	1	1/2	1	1 1/2	8 1/2	3/16	3
1	1 1/2	.....	2	2	10 1/2	5/16	3
1 1/2	2	.....	2 3/8	2 3/8	11 3/4	3/8	3
2	3	.....	3 1/2	3 1/2	16 1/2	7/16	3
3	4	.....	3 1/2	3 1/2	17 3/4	1/2	3

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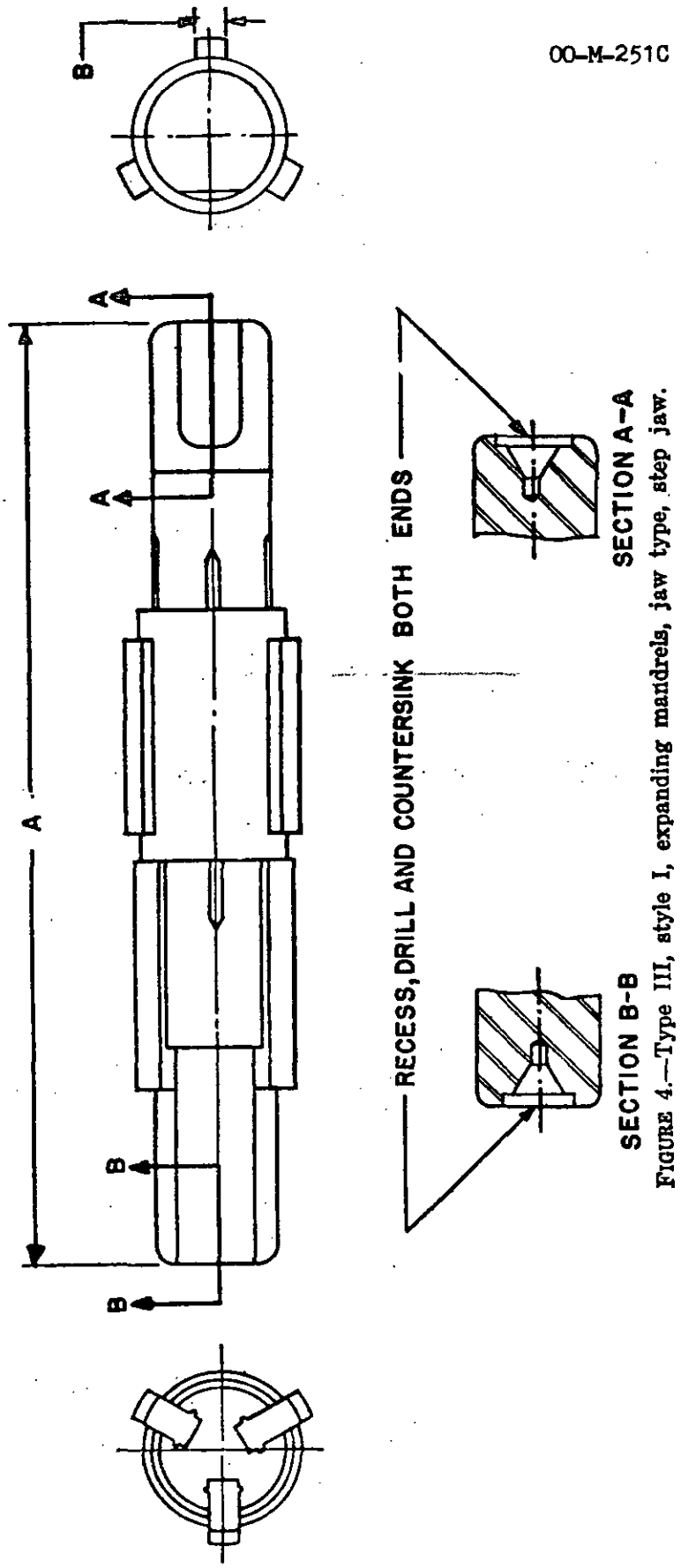


FIGURE 4.—Type III, style I, expanding mandrels, jaw type, step jaw.

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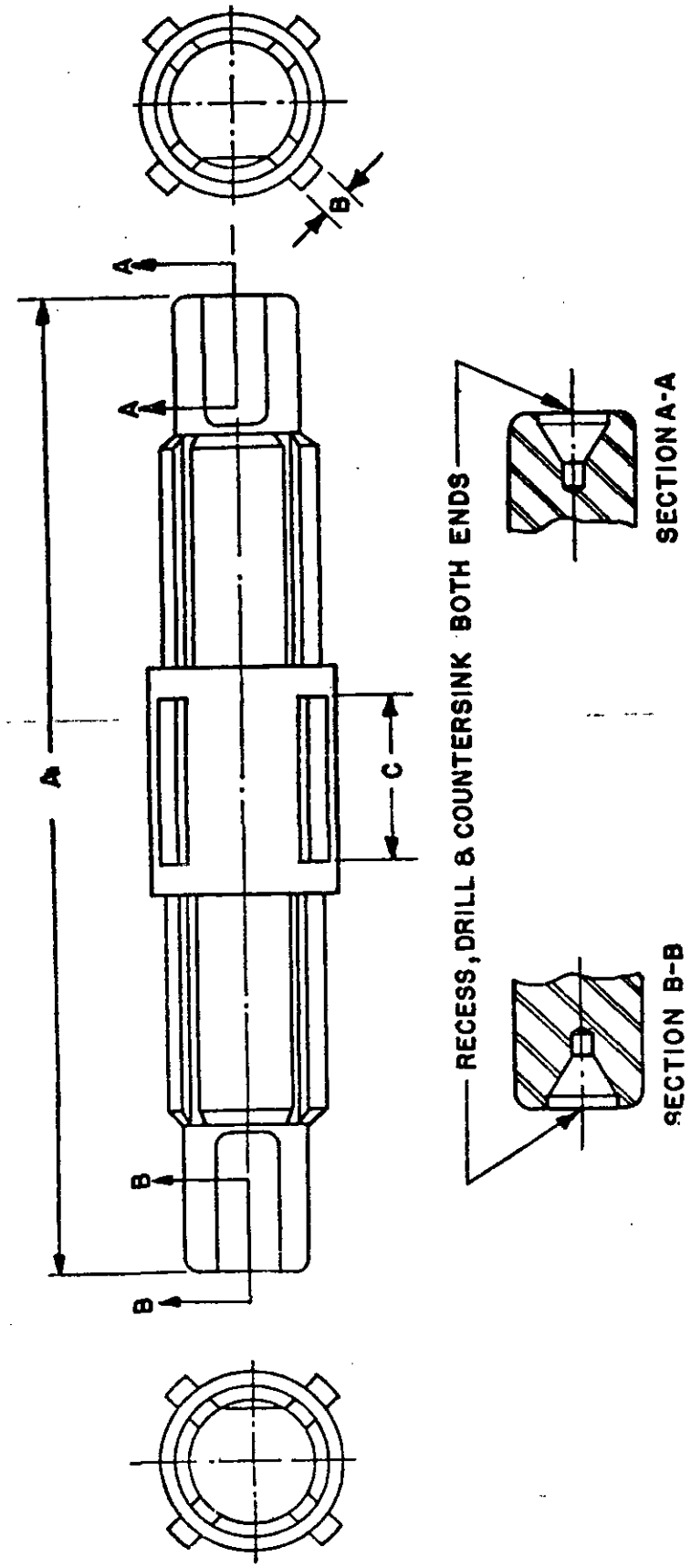


FIGURE 5.—Type III, style II, expanding mandrels, jaw type, straight jaw.

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TABLE IV. Type III, style II, expanding mandrels, jaw type, straight jaw

Range of expansion		Length over-all of arbor	Width of jaws	Length of jaw bearing	Number of jaws per set	Jaws		
Min-imum	Max-imum	A	B	C		Expansion jaw range 1/2 to 6 11/32	Expansion auxiliary jaw range 1 1/8 to 6 11/16	Expansion auxiliary jaw range 3 to 7 1/32
Inches	Inches	Inches	Inches	Inches		Inches	Inches	Inches
1/2	9/16	5	3/16	1	3	1/2 to 9/16		
9/16	21/32	6	3/16	1 1/2	3	9/16 to 21/32		
21/32	3/4	6	3/16	1 1/2	3	21/32 to 3/4		
3/4	7/8	8 1/4	1/4	2	3	3/4 to 7/8		
7/8	1	8 1/4	1/4	2	3	7/8 to 1		
1	1 1/4	9	1/4	2 1/2	4	1 to 1 1/8	1 1/8 to 1 1/4	
1 1/4	1 9/16	11 3/4	5/16	3	4	1 1/4 to 1 13/32	1 13/32 to 1 9/16	
1 9/16	2	13 3/4	3/8	4	4	1 9/16 to 1 25/32	1 25/32 to 2	
2	2 1/2	17 1/4	3/8	5	4	2 to 2 1/4	2 1/4 to 2 1/2	
2 1/2	3 1/4	18 3/4	7/16	5 3/4	4	2 1/2 to 2 3/4	2 3/4 to 3	3 to 3 1/4
3 1/4	4	21	1/2	6	4	3 1/4 to 3 1/2	3 1/2 to 3 3/4	3 3/4 to 4
4	5	23 1/4	5/8	7	4	4 to 4 11/32	4 11/32 to 4 11/16	4 11/16 to 5
5	6	25	5/8	8	4	5 to 5 11/32	5 11/32 to 5 11/16	5 11/16 to 6
6	7 1/32	25	5/8	9	4	6 to 6 11/32	6 11/32 to 6 11/16	6 11/16 to 7 1/32

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3.9.2 Hardness. The jaw-bearing surfaces shall have a hardness of not less than 50 nor more than 61 and the arbors shall be not less than 56 nor more than 58 on the Rockwell "C" scale.

3.9.3 Tolerances and finishes. After hardening and stress relieving, arbor centers shall be accurately ground and lapped to the standard included angle of 60 degrees ± 30 minutes. The jaw slots shall be ground, on dead centers, to a uniform inclined plane and the outside diameter of the arbor finish ground. After assembly, the jaw bearing surfaces shall be finish ground and the surface roughness of the jaw bearings, and the arbor, shall not exceed 16 microinches a.a. and shall conform to B46.1. Type III, Figures 4 and 5, styles I and II expanding jaw mandrels shall have a runout on the jaw bearing surface not exceeding 0.001 inch total indicator reading at any position of the jaws when the mandrel is rotated on snugly adjusted dead centers and with the jaws held solidly against the arbor by ring bushings or other suitable device mounted tightly on the jaws.

3.10 Workmanship. Workmanship of the mandrels shall be of the best quality prevailing among manufacturers of this type of equipment. All components shall be free from imperfections which will adversely affect the general appearance, function, or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).

(b) Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item noted in 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) as follows:

(a) Examination (see 4.3).

(b) Lot acceptance test (see 4.4.1).

(c) Inspection of preparation for delivery (see 4.5).

4.2.2.1 Sampling. When specified (see 6.2), sampling for lot acceptance shall be in accordance with MIL-STD-105 at the inspection level and acceptable quality level (AQL) specified in the contract. Samples for the test shall be selected at random from the items that passed the examination in 4.3.

4.3 Examination. Each production item or sample shall be visually examined to determine compliance with requirements in section 3, using Table V as a guide:

TABLE V. Classification of defects in accordance with MIL-STD-105.

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type and (nominal) size not as specified (type I only).
102	Type, style, and size (range of expansion) not as specified (types II and III only).
103	Material not as required; not free from seams, pipes, cracks, or inclusions.

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Table V. (cont'd)

Categories	Defects
104	Mandrel not solid, one piece construction; not new, evidence of being used or reconditioned (type I only).
105	Mandrel incomplete, component parts or machining operations omitted; not new, evidence of being used, or reconditioned (types II and III only).
106	Mandrel fails to conform to approved plan; incomplete, machining operations omitted, failure of finished ground, lapped, or honed surfaces to "clean up", or evidence of final finishing of working surfaces and centers prior to heat treating operations; evidence of hardening cracks or distortion; mandrel not free from hardening scale (type I only).
107	Mandrel not in accordance with approved plan; arbor not solid, one piece construction; sleeve not one piece, slit and flexible (type II only).
108	Mandrel not in accordance with approved plan or drawing; arbor not solid, one piece construction; jaws not step or straight as applicable; number of jaws per set, steps or sets of jaws per mandrel not as required (type III only).
109	Small diameter (G) and taper per inch of mandrel body not within the required tolerances; not a uniform straight-taper surface (type I only).
110	Finished ground, lapped or honed surfaces fail to "clean up", or evidence of final finishing or working surfaces and centers prior to heat treating operations; evidence of hardening, cracks, or distortion; mandrel not free from hardening scale (types II and III only).
111	Included angle and diameter of centers (N) not within requirements; centers not properly recessed when required (type I only).

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Table V. (cont'd)

Categories	Defects
112	Length of flat (J) and distance across flat (K) on neck diameters not as specified (type I only).
113	Neck diameter end faces not rounded or chamfered as required (type I only).
114	Finished surfaces not free from toolmarks, gouges, nicks, tears, or galled areas; mandrel not free from burs, raised edges, rust, or corrosion (types I, II, and III).
115	Sleeve not properly finished internally to fit arbor and proportioned to provide uniform expansion (type II only).
116	Even number of slits, one-half of which open at each end of sleeve, not provided as designated (type II only).
117	Drilled holes or fillets terminating slits, as applicable, less than the required diameter or radius (type II only).
118	Number of slits not such to provide adequate strength and flexibility or suitable bearing on arbor and in the work hole for all sizes within the applicable range (type II only).
119	Range of expansion of the sleeve diameter not within the designated minimum and maximum limits when properly positioned on arbor; work bearing surfaces not parallel with the axis of arbor centers when located at any position within the applicable range (type II only).
120	Taper per foot, of arbor working surface and sleeve internal working surface, not within the required tolerances; not uniform straight-taper surfaces (type II only).
121	Arbor ends not properly recessed, drilled, and countersunk; included angle of centers not within the designated limits (type II only).

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Table V. (cont'd)

Categories	Defects
122	Flat at neck end for lathe dog, not suitably machined (types II and III only).
123	Evidence of sharp corners at neck diameter and large end of taper, not a smooth fillet radius (when applicable) (types I and III only).
124	Length of arbor and sleeve not in accordance with requirements (type II only).
125	Small end face of arbor not properly chamfered or rounded to prevent damage to sleeve; neck end face not suitably chamfered or rounded (type II only).
126	Jaws fail to fit properly in arbor slots; evidence of binding or excessive play in any free position; slotted sleeve fails to prevent jaws from falling out of mandrel or fails to hold them in proper axial alignment with each other (type III only).
127	Taper of jaws not within requirements; not uniform straight-taper surfaces (type III only).
128	Range of expansion of jaws not within the designated minimum and maximum limits when properly positioned on arbor; work bearing surfaces not parallel with axis of arbor centers when located at any position within the applicable range (type III only).
129	Arbor ends not properly recessed, drilled, and countersunk; included angle of centers not within the required limits (type II only).
130	Evidence of sharp corners at neck diameter and body of arbor, not properly chamfered or radiused (style II only) (type III only).
131	Jaw bearing length of steps not as required, arbor length and width of jaws less than the allowable minimum (style I only) (type III only).
132	Length of jaw bearing, width of jaws and overall length of arbor not in accordance with requirements (style II only) (type III only).

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Table V. (cont'd)

Categories	Defects
133	Arbor end faces not properly chamfered or rounded (type III only).
Minor:	
201	Overall length, neck diameter and length at each end, and drill diameter of centers not within requirements (all types).
202	Marking, nominal size, size and range, manufacturer's name or trademark missing, incorrect, illegible, not permanent, or not located as specified (all types).
203	Preservation, packaging, and packing not as required (all types).

4.4 Test methods.

4.4.1 Lot acceptance test. Each of the sample mandrels selected in accordance with 4.2.2.1 shall be tested as specified in 4.4.2. Any sample mandrel failing the tests shall be rejected and if the number of nonconforming mandrels in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.2 Hardness test. Mandrels shall meet the requirements of 3.7.2, 3.8.2, and 3.9.2, as applicable.

4.4.2.1 Total indicator runout test. Mandrels shall meet the allowable indicator runout specified in 3.7.3, 3.8.3, or 3.9.1, as applicable.

4.4.2.2 Possible test failures. Possible test failures are defined as follows:

(a) Hardness test:

(1) Hardness on all surfaces not within the designated range (type I only).

(2) Hardness of sleeve and arbor on all surfaces not within the designated ranges (type II only).

(3) Hardness of jaw bearing surfaces not within the designated range (type III only).

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(b) Surface finish:

(1) Surface roughness of centers and working surface exceeds the maximum permissible a.a. value (type I only).

(2) Surface roughness of working surfaces of arbor centers and taper, internal and external working surfaces of sleeve, exceeded the maximum permissible a.a. value (type II only).

(3) Surface roughness of working surfaces of arbor and internal and external jaw bearing surfaces exceed the maximum permissible a.a. value (type III only).

(c) Performance test:

(1) Total indicator reading exceeds the maximum allowable limit at any position along the working surface of the sleeve for any position of the sleeve on the arbor, within the applicable size range, when rotated on snugly adjusted dead centers and with the sleeve held solidly against the arbor by ring bushings or other suitable device mounted tightly on sleeve (types II and III only).

(2) Mandrel fails to tighten or release as required (types II and III only).

(3) Total indicator reading at any position along the working surface when rotated on snugly adjusted dead centers exceeds the maximum allowable limit (type I only).

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the mandrels shall be inspected to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be in accordance with PPP-T-1150. The levels of preservation, packaging, and packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The mandrels covered by this specification are intended for use in machine shop facilities for various machining and inspection operations.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type, style, and size mandrel required (see 1.2.1 and 1.2.2).

(c) First article, when required (see 3.1).

(d) Size and quantity of repair sleeves, type II mandrel, when required (see 3.8.4).

(e) First article inspection, if different (see 4.2.1).

(f) Quality conformance inspection, if different (see 4.2.2).

(g) Sampling, as required (see 4.2.2.1).

(h) Preservation, packaging, packing, and marking if different (see 5.1).

(i) Selection of applicable level of packaging and packing (see 5.1).

MILITARY INTEREST:

Custodians:
Army - WC
Navy - YD
Air Force - 84

Review activities:
Army - none
Navy - none
Air Force - none

User activities:
Army - SM, ME
Navy - none
Air Force - none

Preparing activity:
Army - WC

CIVIL AGENCY INTEREST:
GSA-FSS

Project No. 3460-0063

(2448)

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein. Price 15 cents each.

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PLATE NO. 15419

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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